

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407793
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Tamagawa; Shuichi

Display control device and display control system

Abstract

A display control device is configured to control image display of a display device for remote operation of a moving body. The display device includes one or more screens respectively configured to display one or more intersection images captured by one or more cameras. The one or more cameras are installed at an intersection through which the moving body passes and are configured to each image a traffic situation in an area of the intersection. The display control device includes a processor configured to execute an image process of arranging the one or more intersection images on the one or more screens in different modes depending on a relation between one or more imaging directions of the one or more cameras and a traveling direction of the moving body.

Inventors:	Tamagawa; Shuichi (Kitanagoya, JP)
Applicant:	TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)
Family ID:	91953156
Assignee:	TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)
Appl. No.:	18/519606
Filed:	November 27, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240251059 A1	Jul. 25, 2024

Foreign Application Priority Data

JP	2023-009490	Jan. 25, 2023
----	-------------	---------------

Publication Classification

Int. Cl.: H04N7/18 (20060101); G06T3/60 (20240101); H04N23/90 (20230101)

U.S. Cl.:

CPC H04N7/181 (20130101); G06T3/60 (20130101); H04N23/90 (20230101);

Field of Classification Search

CPC: G05D (1/0033); G05D (1/0246); G08G (1/0116); H04N (23/90); H04N (7/181); G06T (3/60)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7777616	12/2009	Mase	382/104	G08G 1/164
8406977	12/2012	Tamaoki	701/96	B60W 30/18154
11544918	12/2022	Kuybeda	N/A	G06V 10/143
2007/0219709	12/2006	Ikeda	701/516	G02B 27/01
2020/0310409	12/2019	Otaka	N/A	N/A
2022/0207279	12/2021	Kuybeda	N/A	H04N 25/671

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H09180087	12/1996	JP	N/A
2004021514	12/2003	JP	N/A
2020167550	12/2019	JP	N/A

Primary Examiner: Tchoussou; Boubacar Abdou

Attorney, Agent or Firm: Dinsmore & Shohl LLP

Background/Summary

CROSS-REFERENCES TO RELATED APPLICATION

(1) The present disclosure claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2023-009490, filed on Jan. 25, 2023, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a display control device and a display control system for remote operation of a moving body.

BACKGROUND

(3) JP 2020-167550 A discloses a communication device configured to acquire information used for remote operation of a moving body. The communication device obtains information from a road management camera based on the geographic location of the moving body. The road management camera is installed at a traffic signal or beside a road and captures an image of the road.

(4) JP H09-180087 A discloses a traffic information providing system configured to capture an image of a road condition and display the captured image on an in-vehicle display device. Further, JP 2004-021514 A discloses a traffic information display system including an information display

device mounted on a vehicle. This information display device obtains the course direction of the subject vehicle in the received captured image from map information of a selected point and a received camera parameter, and combines the information indicating the course direction with the captured image to display the combined image on a screen.

SUMMARY

(5) In order to confirm safety of a remote operator during remote operation (remote driving, remote assistance, or remote monitoring) of a moving body, such as an automobile, it is conceivable to display, on a screen of a display device of a remote operator terminal, an image of a camera installed at an intersection. In order to display this kind of image, it is required that the remote operator who views the image displayed on the display device can easily grasp the traffic situation while instantaneously grasping the traveling direction of the moving body in the image.

(6) However, the technique described in JP 2020-167550 A relates to acquisition of an image from the road management camera, and does not disclose how the acquired image of the intersection should be displayed on the display device of the remote operator terminal from the viewpoint of improving the visibility of the remote operator. This also applies to the techniques described in JP H09-180087 A and JP 2004-021514 A.

(7) The present disclosure has been made in view of the problem described above, and an object thereof is to provide a display control device and a display control system that can improve the visibility of a remote operator with respect to image display of a display device for remote operation of a moving body.

(8) A display control device according to the present disclosure is configured to control image display of a display device for remote operation of a moving body. The display device includes one or more screens respectively configured to display one or more intersection images captured by one or more cameras. The one or more cameras are installed at an intersection through which the moving body passes and are configured to each image a traffic situation in an area of the intersection. The display control device includes a processor configured to execute an image process of arranging the one or more intersection images on the one or more screens in different modes depending on a relation between one or more imaging directions of the one or more cameras and a traveling direction of the moving body.

(9) According to the present disclosure, the visibility of the remote operator with respect to the image display of the display device for the remote operation of the moving body can be improved.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram schematically illustrating a configuration example of a remote operation system to which a display control device according to an embodiment is applied;

(2) FIG. 2 is a diagram showing an example of a screen configuration of a display device shown in FIG. 1;

(3) FIG. 3A is a diagram used to describe an image process IP applied to an intersection P1 in which two intersection cameras are installed;

(4) FIG. 3B is a diagram used to describe the image process IP applied to the intersection P1 in which two intersection cameras are installed;

(5) FIG. 4A is a diagram used to describe the image process IP applied to the intersection P1;

(6) FIG. 4B is a diagram used to describe the image process IP applied to the intersection P1;

(7) FIG. 4C is a diagram used to describe the image process IP applied to the intersection P1;

(8) FIG. 5A is a diagram used to describe an image process IP applied to an intersection P2 in which only one intersection camera is installed;

(9) FIG. 5B is a diagram used to describe the image process IP applied to the intersection P2; and

(10) FIG. 6 is a flowchart illustrating an example of a flow of the image process IP according to an embodiment.

DETAILED DESCRIPTION

(11) Hereinafter, embodiments of the present disclosure will be described with reference to the accompanying drawings. In addition, common elements in the drawings are denoted by the same reference numerals, and redundant description thereof will be omitted or simplified.

1. Configuration of Remote Operation System

(12) FIG. 1 is a diagram schematically illustrating a configuration example of a remote operation system **1** to which a display control device according to an embodiment is applied. The remote operation system **1** includes a vehicle **10**, a remote operator terminal **20**, a camera device **30**, and a management server **40**.

1-1. Vehicle

(13) The vehicle **10** is an example of a “moving body” that is a target of “remote operation” according to the present disclosure, and more specifically, is, for example, an automobile. The vehicle **10** includes a global navigation satellite system (GNSS) receiver **11**, a plurality of in-vehicle cameras **12**, a travel device **13**, a communication device **14**, an antenna **15**, and a control device **16**.

(14) The GNSS receiver **11** is used to acquire a position and an orientation of the vehicle **10**. The plurality of in-vehicle cameras **12** recognize (detect) a situation around the vehicle **10**. The plurality of in-vehicle cameras **12** include, for example, cameras that respectively image the front center, the left front, and the right front of the vehicle **10**.

(15) The travel device **13** includes a drive device, a brake device, and a steering device. The drive device includes, for example, at least one of an electric motor and an internal combustion engine for driving the vehicle **10**. The brake device includes a brake actuator for braking the vehicle **10**. The steering device includes, for example, an electric motor configured to steer wheels.

(16) The communication device **14** communicates with the outside of the vehicle **10**. Specifically, the communication device **14** communicates with, for example, the remote operator terminal **20** via the antenna **15**.

(17) The control device **16** is a computer configured to control the vehicle **10**. The control device **16** includes one or more processors and one or more memory devices. The one or more processors execute various processes related to control of the vehicle **10**. The one or more memory devices store various types of information necessary for the processes executed by the one or more processors.

(18) During the remote operation of the vehicle **10**, the control device **16** communicates with the remote operator terminal **20** via the communication device **14**. The control device **16** transmits vehicle information VCL to the remote operator terminal **20**. The vehicle information VCL is information necessary for the remote operation by the remote operator, and includes surrounding situation information of the vehicle **10**. The surrounding situation information includes images captured by the in-vehicle cameras **12**. The surrounding situation information may include sound detected by an in-vehicle microphone. The surrounding situation information may include object information related to a moving object around the vehicle **10**. The object information indicates a relative position and a relative speed of an object with respect to the vehicle **10**. Moreover, the vehicle information VCL may include vehicle state information, such as a speed, an acceleration, a yaw rate, and a steering angle of the vehicle **10**, and vehicle position information.

(19) Further, the control device **16** receives remote operation information from the remote operator terminal **20**. The remote operation information is related to the remote operation by the remote operator. For example, the remote operation information includes the amount of operation by the remote operator. The control device **16** performs vehicle traveling control by controlling the travel device **13** in accordance with the received remote operation information.

1-2. Remote Operator Terminal

(20) The remote operator terminal **20** is a terminal device used when the remote operator remotely operates the vehicle **10**. The “remote operation” according to the present disclosure is a concept including “remote driving” in which a remote operator drives a moving body from a remote place, “remote assistance” in which a remote operator assists the driving of a moving body from a remote place, and “remote monitoring” in which a remote operator monitors the driving of a moving body from a remote place. In addition, when the remote operation is performed as the remote driving, the remote operator terminal **20** corresponds to a remote cockpit.

(21) The remote operator terminal **20** includes a communication device **21**, an antenna **22**, a display device **23**, a speaker **24**, an operating device **25**, and a control device **26**.

(22) The communication device **21** performs communication (wireless communication) with the vehicle **10** via the antenna **22**. Further, the communication device **21** communicates with the management server **40**.

(23) The display device **23** presents various kinds of information to the remote operator by displaying the various kinds of information. The various kinds of information include images of intersection areas (intersection images) captured by a plurality of intersection cameras **31**. That is, each of the intersection images is an image indicating the traffic situation in an area of the intersection into which the vehicle **10** enters.

(24) FIG. 2 is a diagram showing an example of a screen configuration of the display device **23** shown in FIG. 1, and shows the display device **23** viewed from the remote operator. That is, the right side of the drawing sheet corresponds to the right side of the screen when viewed from the remote operator, and the left side of the drawing sheet corresponds to the left side of the screen when viewed from the remote operator.

(25) In the example illustrated in FIG. 2, the display device **23** includes five monitors **231**, **232**, **233**, **234**, and **235**. The monitors **231**, **232**, and **233** display images in front of the vehicle **10** (vehicle front images) captured by the plurality of in-vehicle cameras **12**. More specifically, as an example, one or more vehicle front images include a front center image displayed by the monitor **231**, a left front image displayed by the monitor **232**, and a right front image displayed by the monitor **233**. As shown in FIG. 2, the monitor **232** is arranged on the left side of the monitor **231**, and the monitor **233** is arranged on the right side of the monitor **231**.

(26) The monitors **234** and **235** each display an intersection image. As shown in FIG. 2, the monitors **234** and **235** are arranged side by side in the left-right direction, and are arranged above the monitors **231** to **233**, for example. Specific examples of images displayed on the monitors **234** and **235** will be described below with reference to FIGS. 3A, 3B, 4A, 4B, 4C, 5A, and 5B. In addition, in the example illustrated in FIG. 2, the display device **23** includes one monitor (each of the monitors **231** to **235**) for one image. Instead of this example, the “display device” according to the present disclosure may include one monitor configured display both the vehicle front image and the intersection image.

(27) The speaker **24** presents sound information to the remote operator by outputting the sound information. The sound information includes the sound detected by the in-vehicle microphone described above.

(28) The operating device **25** receives an input from the remote operator. For example, the operating device **25** includes one or more remote operation members operated by the remote operator when remotely operating the vehicle **10**. The one or more remote operation members include an accelerator pedal, a brake pedal, and a steering wheel. In addition, the operating device **25** includes switches for other various operations related to the remote operation.

(29) The control device **26** controls the remote operator terminal **20**. The control device **26** includes one or more processors **27** (hereinafter, simply referred to as “processor **27**”) and one or more memory devices **28** (hereinafter, simply referred to as “memory device **28**”). The processor **27** executes various processes. For example, the processor **27** includes a central processing unit (CPU). The memory device **28** stores various types of information necessary for the processes

executed by the processor **27**. The memory device **28** is, for example, at least one of a volatile memory, a nonvolatile memory, a hard disk drive (HDD), and a solid state drive (SSD).

(30) Remote operation program is a computer program executed by the processor **27**. The function of the control device **26** is realized by the processor **27** executing the remote operation program. The remote operation program is stored in the memory device **28**. Alternatively, the remote operation program may be recorded in a computer-readable recording medium. The remote operation program may be provided via a network.

(31) The control device **26** communicates with the vehicle **10** via the communication device **21**. The control device **26** receives the vehicle information VCL transmitted from the vehicle **10**. The control device **26** presents the vehicle information VCL to the remote operator via the display device **23** and the speaker **24**. For example, the control device **26** displays the images captured by the in-vehicle camera **12** on the display device **23**. Further, the control device **26** outputs the sound detected by the in-vehicle microphone from the speaker **24**. The remote operator can recognize the state of the vehicle **10** and the surrounding situation based on the vehicle information VCL.

(32) The remote operator operates the remote operation members of the operating device **25**. The operating amounts of the remote operation members are detected by sensors provided in the remote operation members. The control device **26** generates the remote operation information reflecting the operating amounts of the remote operation members by the remote operator. Then, the control device **26** transmits the remote operation information to the vehicle **10** via the communication device **21**.

1-3. Intersection Camera

(33) The plurality of intersection cameras **31** (hereinafter, also simply referred to as “camera **31**”) are installed at each intersection through which the vehicle **10** passes, and capture an image of an intersection area (more specifically, the inside of the intersection and the vicinity thereof). Each of the plurality of cameras **31** is configured as, for example, the camera device **30** including the camera **31** and a communication device **32**. More specifically, for example, one or more cameras **31** are installed at one intersection. In the example shown in FIG. **3** described below, two cameras **31** are installed for one intersection.

(34) An image of an intersection area (intersection image) captured by each camera **31** is transmitted to the management server **40** via the communication device **32**. The management server **40** is configured to be able to communicate with each camera device **30**, and manages the intersection image transmitted from each camera device **30**. The management server **40** is also configured to be able to communicate with the remote operator terminal **20**. When the vehicle **10** approaches an intersection area, the control device **26** of the remote operator terminal **20** requests the management server **40** to transmit an image of the intersection area. In response to the request, the management server **40** transmits the image of the corresponding intersection area to the remote operator terminal **20**.

(35) Additionally, communication between the management server **40** and each camera device **30** may be either wired communication or wireless communication. This also applies to communication between the management server **40** and the remote operator terminal **20**. Furthermore, unlike the example illustrated in FIG. **1**, the acquisition of the intersection image by the remote operator terminal **20** may be performed by direct communication with each camera device **30** without going through the management server **40**.

2. Control of Image Display on Display Device

(36) In the remote operation system **1** described above, the control device **26** of the remote operator terminal **20** controls the image display of the display device **23** for the remote operation of the vehicle **10** (moving body). Therefore, the control device **26** corresponds to an example of the “display control device” according to the present disclosure, and a system that includes the control device **26** and the intersection camera **31** corresponds to an example of the “display control system” according to the present disclosure.

(37) When the image of the intersection camera **31** is displayed on the display device **23** for the safety confirmation by the remote operator, it is required for the display of the image requires that the remote operator viewing the image displayed on the display device **23** can easily grasp the traffic situation while instantaneously grasping a traveling direction D_v of the vehicle **10** in the image.

(38) More specifically, as illustrated in FIG. 2, on the screen of the display device **23** installed in the remote operator terminal **20**, the image from the in-vehicle camera **12** (vehicle front image) and the image from the intersection camera **31** (intersection image) are simultaneously displayed and mixed. The remote operator is required to view all of these multiple images. Also, the remote operator recognizes the space in the image and converts the space in his or her brain as necessary for the space recognition every time he or she puts his or her line of sight on each image. Moreover, the orientation of the intersection camera **31** (imaging direction D_c) and the installation location thereof vary depending on the environment of each intersection. Therefore, the intersection image of each intersection is affected by the differences in the orientation and the installation location of the intersection camera **31**. Therefore, it is difficult to instantaneously determine the positional relation between a moving object, such as another vehicle or a pedestrian, in an intersection area and the vehicle **10** and instantaneously determine the moving direction of the moving object (for example, whether or not the moving object is approaching the vehicle **10**) while instantaneously grasping the traveling direction D_v of the vehicle **10** in the intersection image.

(39) In view of the above issue, in the present embodiment, the control device **26** (processor **27**) executes the following “image process IP”. According to the image process IP, the intersection image is arranged on the screen of the display device **23** in different modes depending on the relation between the imaging direction D_c of the camera **31** and the traveling direction D_v of the vehicle **10** (moving body).

(40) A specific example of the image process IP will be described with reference to FIGS. 3A, 3B, 4A, 4B, and 4C. As shown in FIGS. 3A and 4A, this specific example will be described by taking, as an example, an intersection **P1** where two intersection cameras **31A** and **31B** (a first camera **31A** and a second camera **31B**) are installed.

(41) The area of the intersection **P1** imaged by the cameras **31A** and **31B** includes not only an area located in the intersection **P1** but also a first road **R1** on which the vehicle **10** travels and a second road **R2** that intersects with the first road **R1** at the intersection **P1**.

(42) More specifically, FIG. 3A illustrates a traffic situation in the area of the intersection **P1** when the vehicle **10** traveling on the first road **R1** along a traveling direction D_{v1} is approaching the intersection **P1**. In this situation, the first camera **31A** is located beside the first road **R1** on the right side of the vehicle **10** when viewed from the vehicle **10**. Also, the first camera **31A** captures not only the area in the intersection **P1** but also an area of the second road **R2** located on the left side of the intersection **P1**. Another vehicle **2** traveling on the second road **R2** is included in an imaging area **A1** of this first camera **31A**. In addition, instead of the described above, the installation location of the first camera **31A** that captures the area of the second road **R2** on the left side of the intersection **P1** illustrated in FIG. 3A may be, for example, inside the intersection **P1** or beside the road on the left side of the vehicle **10**.

(43) Moreover, in the above-described situation shown in FIG. 3A, the second camera **31B** is located beside the first road **R1** on the left side of the vehicle **10** when viewed from the vehicle **10**. Also, the second camera **31B** captures not only the area in the intersection **P1** but also an area of the second road **R2** located on the right side of the intersection **P1**. A pedestrian **3** crossing the second road **R2** is included in an imaging area **A2** of this second camera **31B**. In addition, instead of the example described above, the installation location of the second camera **31B** that captures the area of the second road **R2** on the right side of the intersection **P1** illustrated in FIG. 3A may be, for example, inside the intersection **P1** or beside the road on the right side of the vehicle **10**.

(44) FIG. 3B shows an image display pattern **PT1** obtained by the image process IP according to

the embodiment. The image display pattern PT1 corresponds to the display of the intersection image on the display device 23 in the above-described situation shown in FIG. 3A. The two intersection images shown in FIG. 3B correspond to the images themselves displayed on the screen of the display device 23 as it is in a state of being captured by the first and second cameras 31A and 31B, respectively. In addition, various arrows shown in the intersection images in FIG. 3B are shown for reference of description, and are not included in the intersection images. The same applies to FIGS. 4B, 4C, 5A, and 5B described below.

(45) In the situation shown in FIG. 3A, directions Dc1' and Dc2' of vehicle-traveling-direction components of vectors that indicate respective imaging directions Dc1 and Dc2 of the first and second cameras 31A and 31B coincide with a traveling direction Dv1. In this situation, the control device 26 (processor 27) selects the image display pattern PT1 in the image process IP.

(46) In the image display pattern PT1, as shown in FIG. 3B, the image of the first camera 31A (i.e., the intersection camera capturing the area of the second road R2 located on the “left side” of the intersection P1 when viewed from the vehicle 10) is arranged on the “left side” in the screen of the display device 23. More specifically, the image of the first camera 31A is displayed on the screen of the left-side monitor 234. On the other hand, the image of the second camera 31B (i.e., the intersection camera capturing the area of the second road R2 located on the “right side” of the intersection P1 when viewed from the vehicle 10) is arranged on the “right side” in the screen of the display device 23. More specifically, the image of the second camera 31B is displayed on the screen of the right-side monitor 235. As described above, according to the image display pattern PT1, the image of the first camera 31A is arranged on the left side of the image of the second camera 31B.

(47) According to the image display pattern PT1, another vehicle 2 that is present on the left side (more specifically, the left front side) of the vehicle 10 at the actual intersection P1 and is traveling in a direction away from the vehicle 10 can be displayed on the screen of the monitor 234 on the left side when viewed from the remote operator so as to be seen in the same manner as the actual appearance. Therefore, the intersection image of the first camera 31A can be displayed so as to have the same appearance as those of the vehicle front images displayed on the monitors 231 and 232.

(48) Moreover, according to the image display pattern PT1, the pedestrian 3 who is present on the right side (more specifically, the right front side) of the vehicle 10 at the actual intersection P1 and is crossing the second road R2 in a direction approaching the vehicle 10 can be displayed on the screen of the monitor 235 on the right side when viewed from the remote operator so as to be seen in the same manner as the actual appearance. Therefore, the intersection image of the second camera 31B can be displayed so as to have the same appearance as those of the vehicle front images displayed on the monitors 231 and 233.

(49) FIG. 4B is a diagram used to describe an issue that occurs when the image display pattern PT by the image process IP is not changed in accordance with the traveling direction Dv of the vehicle 10.

(50) FIG. 4A shows a traffic situation in the area of the intersection P1 when the vehicle 10 traveling along a traveling direction Dv2 opposite to the traveling direction Dv1 shown in FIG. 3A is approaching the intersection P1. FIG. 4B corresponds to a comparative example in which, in the situation shown in FIG. 4A, the image of the first camera 31A is displayed as it is on the screen of the left-side monitor 234 and the image of the second camera 31B is displayed as it is on the screen of the right-side monitor 235 without special consideration, similarly to the example shown in FIG. 3B. In addition, FIG. 4B shows the intersection images at the same time point as the intersection images shown in FIG. 3B.

(51) According to the image display shown in FIG. 4B, another vehicle 2 that is present in the right front of the vehicle 10 at the actual intersection P1 as shown in FIG. 4A is displayed on the left side when viewed from the remote operator. Also, the pedestrian 3 that is present in the left front of the

vehicle **10** at the actual intersection **P1** is displayed on the right side when viewed from the remote operator. More specifically, on the left-side monitors **232** and **234** (see FIG. 2) of the display device **23**, the intersection image of the area actually located on the right side of the vehicle **10** is arranged together with the left front image of the vehicle **10**. Similarly, on the right-side monitors **233** and **235**, the intersection image of the area actually located on the left side of the vehicle **10** is arranged together with the right front image of the vehicle **10**. This results in confusion for the remote operator.

(52) FIG. 4C shows an image display pattern **PT2** obtained by the image process **IP** according to the embodiment. In view of the issue described with reference to FIG. 4B, according to the image process **IP**, the image display pattern **PT2** is selected in order to display the intersection images on the display device **23** under the above-described situation shown in FIG. 4A. In addition, FIG. 4C shows intersection images at the same time point as the intersection images shown in FIGS. 3B and 4B.

(53) To be specific, in the situation shown in FIG. 4A, the directions **Dc1'** and **Dc2'** of the vehicle-traveling-direction components of the vectors that indicate the respective imaging directions **Dc1** and **Dc2** of the first and second cameras **31A** and **31B** face the traveling direction **Dv2**. Therefore, the control device **26** (processor **27**) selects the image display pattern **PT2** in the image process **IP**.

(54) In the image display pattern **PT2**, as shown in FIG. 4C, the image of the first camera **31A** (i.e., the intersection camera capturing the area of the second road **R2** located on the “right side” of the intersection **P1** when viewed from the vehicle **10**) is arranged on the “right side” in the screen of the display device **23**. More specifically, the image of the first camera **31A** is displayed on the screen of the right-side monitor **235**. On the other hand, the image of the second camera **31B** (i.e., the intersection camera capturing the area of the second road **R2** located on the “left side” of the intersection **P1** when viewed from the vehicle **10**) is arranged on the “left side” in the screen of the display device **23**. More specifically, the image of the second camera **31B** is displayed on the screen of the left-side monitor **234**.

(55) As described above, according to the image display pattern **PT2**, the image of the first camera **31A** is arranged on the right side of the image of the second camera **31B**. As a result, in consideration of the fact that the relation with the imaging direction **Dc** of the camera **31** is different between the traveling direction **Dv1** and the traveling direction **Dv2**, the arrangement laterally opposite to the arrangement of the two intersection images of the image display pattern **PT1** is adopted. Thus, the two intersection images can be arranged so as to be matched with the left and right of the vehicle front images displayed on the display device **23**.

(56) However, with only the measures described above, the horizontal direction in each intersection image becomes opposite to the actual horizontal direction. More specifically, the moving direction of the moving object (for example, another vehicle **2** illustrated in FIG. 4A) moving along the second road **R2** becomes opposite to the actual moving direction and the moving direction of the vehicle **2** displayed in the vehicle front image (right front image). As a result, in this example of the vehicle **2**, the remote operator may erroneously determine that the vehicle **2** is approaching the vehicle **10** even though the vehicle **2** is actually moving away from the vehicle **10**.

(57) Therefore, according to the image display pattern **PT2**, as shown in FIG. 4C, each of the images of the first and second cameras **31A** and **31B** is flipped horizontally. Accordingly, even when the above-described directions **Dc1'** and **Dc2'** related to the first and second cameras **31A** and **31B** face the traveling direction **Dv2**, the intersection images can be appropriately displayed such that the confusion of the remote operator is reduced.

(58) Next, FIGS. 5A and 5B are diagrams used to describe an image process **IP** applied to an intersection **P2** in which only one intersection camera **31** (hereinafter, referred to as a camera **31C** for convenience) is installed. The intersection image shown in FIG. 5A corresponds to the image itself displayed on the screen of the display device **23** as it is in a state of being captured by the camera **31C**. In addition, when two screens for displaying the intersection images are provided as

in the example of the display device **23** shown in FIG. 2, the image captured by the camera **31C** is displayed on any one of the two screens. Further, the “display device” to which the example shown in FIGS. 5A and 5B is applied may include only one screen for displaying one intersection image.

(59) FIG. 5A also shows a traffic situation in an area of the intersection **P2** when the vehicle **10** traveling along a traveling direction **Dv3** is approaching the intersection **P2**. In this situation, the camera **31C** is located beside a road on the right side of the vehicle **10** when viewed from the vehicle **10**. Also, the camera **31C** captures not only the area in the intersection **P2** but also areas of a first road **R3** and a second road **R4** that intersect with each other at the intersection **P2**. Further, the image (i.e., the imaging area) of this camera **31C** includes a pedestrian **4** crossing a crosswalk.

(60) In the situation shown in FIG. 5A, a direction **Dc3'** of the vehicle-traveling-direction component of a vector that indicates an imaging direction **Dc3** of the camera **31C** coincides with a traveling direction **Dv3**. In this situation, according to the image process **IP**, the image of the camera **31C** is displayed on the screen as shown in FIG. 5A as it is without being flipped horizontally.

(61) On the other hand, in a situation where the vehicle **10** traveling along a traveling direction **Dv4** (see FIG. 5A) opposite to the traveling direction **Dv3** approaches the intersection **P2**, the direction **Dc3'** of the vehicle-traveling-direction component of the vector that indicates the imaging direction **Dc3** of the camera **31C** faces the traveling direction **Dv4**. In this situation, according to the image process **IP**, the image acquired from the camera **31C** is displayed on the screen as shown in FIG. 5B by being flipped horizontally.

(62) As described above, according to the image process **IP** applied to the example shown in FIGS. 5A and 5B, even when one camera **31C** is installed for one intersection **P2**, the intersection image is displayed on the screen in different modes depending on the relation between the imaging direction **Dc3** of the camera **31C** and the traveling direction **Dv**.

(63) To be more specific, according to the image process **IP** applied to the example shown in FIGS. 5A and 5B, similarly to the example shown in FIGS. 3B and 4C, the intersection image (see FIG. 5B) displayed on the screen when the vehicle **10** enters the intersection **P2** from the traveling direction **Dv4** (second direction) opposite to the traveling direction **Dv3** is flipped horizontally with respect to the intersection image (see FIG. 5A) displayed on the screen when the vehicle **10** enters the intersection **P2** from the traveling direction **Dv3** (first direction).

(64) As a result, when the traveling direction **Dv** is **Dv3**, the pedestrian **4** displayed on the left side (left front side) of the vehicle **10** in the vehicle front image showing the actual traffic situation can be displayed on the left side similarly in the intersection image as shown in FIG. 5A. Further, when the traveling direction **Dv** is **Dv4**, the pedestrian **4** displayed on the right side (right front side) of the vehicle **10** in the vehicle front image showing the actual traffic situation can be displayed on the right side similarly in the intersection image as shown in FIG. 5B. That is, regardless of whether the traveling direction **Dv** is **Dv3** or **Dv4**, the occurrence of a difference between the vehicle front image (i.e., the actual situation) and the intersection image can be prevented as to on which side of the vehicle **10** a moving object (for example, the pedestrian **4**) in the intersection area is located. As described above, according to the image process **IP**, even in the example in which one camera **31C** is installed for one intersection **P2**, it is possible to appropriately display the intersection image so as to reduce the confusion of the remote operator caused by the difference in the traveling direction **Dv** with respect to the imaging direction **Dc3** of the camera **31C**.

(65) FIG. 6 is a flowchart illustrating an example of a flow of the image process **IP** according to the embodiment. The processing of this flowchart is started when the remote operation of the vehicle **10** is started, and is ended when the remote operation is ended.

(66) In step **S100**, the control device **26** (processor **27**) determines whether or not the vehicle **10** has approached an intersection where the intersection camera **31** is installed, based on, for example, the vehicle position information transmitted from the vehicle **10**. As a result, when this determination result is Yes, the processing proceeds to step **S102**.

(67) In step **S102**, the control device **26** determines whether the imaging direction **Dc** of the camera **31** coincides with or faces the traveling direction **Dv** of the vehicle **10**. More specifically, the direction **Dc'** of the vehicle-traveling-direction component of the vector that indicates the imaging direction **Dc** is compared with the traveling direction **Dv**. The information on the imaging direction **Dc** used for this determination is acquired from, for example, the camera device **30** together with the intersection image. Then, the control device **26** calculates the direction **Dc'** from the acquired imaging direction **Dc**. Further, the control device **26** calculates the traveling direction **Dv** based on, for example, the vehicle position information.

(68) As a result, when the imaging direction **Dc** (direction **Dc'**) coincides with the traveling direction **Dv** (step **S102**; Yes), the processing proceeds to step **S104**. In step **S104**, the control device **26** displays the intersection image acquired from the intersection camera **31** at a designated cycle on the screen of the display device **23** as it is without flipping the intersection image horizontally. More specifically, in an example in which the two intersection images from the two cameras **31** are displayed on the two screens, respectively, the two intersection images are displayed as in the image display pattern **PT1** shown in FIG. 3B.

(69) On the other hand, when the imaging direction **Dc** (direction **Dc'**) faces the traveling direction **Dv** (step **S102**; No), the processing proceeds to step **S106**. In step **S106**, the control device **26** determines whether or not there are two intersection images to be displayed.

(70) When there are two intersection images in step **S106**, the processing proceeds to step **S108**. In step **S108**, the control device **26** displays two intersection images on the two screens, respectively, as in the image display pattern **PT2** shown in FIG. 4C. To be more specific, the arrangement of the two intersection images on the screen of the display device **23** is performed so as to be laterally (or horizontally) opposite to the arrangement of the two intersection images by the processing of step **S104**. Further, the two intersection images are flipped horizontally with respect to the images acquired from the cameras **31**.

(71) On the other hand, when the number of intersection images to be displayed is one (step **S106**; No), the processing proceeds to step **S110**. In step **S110**, the control device **26** displays the intersection image on the screen while horizontally flipping the image acquired from the camera **31**.

(72) In step **S112** subsequent to the processing of step **S104**, **S108**, or **S110**, the control device **26** determines whether or not the vehicle **10** has passed through the intersection (more specifically, the intersection subject to the determination in step **S100**) based on, for example, the vehicle position information.

(73) As a result, while the vehicle **10** has not yet passed through the intersection described above, the display of the intersection images by the processing of step **S104**, **S108**, or **S110** is continued. On the other hand, after the vehicle **10** has passed through the intersection, the control device **26** proceeds to step **S114** and hides the intersection image corresponding to the intersection. As a result, the images displayed on the display device **23** can be minimized in accordance with the current traveling point of the vehicle **10**. That is, the remote operator can concentrate on the vehicle front images until the vehicle **10** approaches the next intersection.

(74) As described above, according to the image process **IP** of the present embodiment, one or more intersection images are arranged on the screen of the display device **23** in different modes depending on the relation between one or more imaging directions **Dc** of one or more cameras **31** and the traveling direction **Dv** of the vehicle **10** (moving body). Thus, the visibility of the remote operator with respect to the image display of the display device **23** can be improved. More specifically, the remote operator can intuitively determine the position and the moving direction of the moving object existing in the intersection area with respect to the vehicle **10** based on the one or more intersection images displayed on the display device **23**.

Claims

1. A display control device for controlling image display of a display device for remote operation of a moving body, the display device including one or more screens respectively configured to display one or more intersection images captured by one or more cameras, the one or more cameras being installed at an intersection through which the moving body passes and each being configured to image a traffic situation in an area of the intersection, the display control device comprising: a processor configured to execute an image process of arranging the one or more intersection images on the one or more screens in different modes depending on a relation between one or more imaging directions of the one or more cameras and a traveling direction of the moving body, wherein the one or more cameras include first and second cameras, an area of the intersection captured by the first and second cameras includes a second road that intersects, at the intersection, a first road on which the moving body travels, when a direction of a vehicle-traveling-direction component of a vector that indicates an imaging direction of each of the first and second cameras coincides with the traveling direction, the processor is configured, in the image process, to arrange, on a left side of an image of the second camera capturing a first area of the second road located on a right side of the intersection when viewed from the moving body, an image of the first camera capturing a second area of the second road located on a left side of the intersection, the processor is configured to compare the direction of the vehicle-traveling-direction-component of the vector that indicates the imaging direction with the traveling direction, and calculate the traveling direction based on vehicle position information, the processor is configured to display the one or more intersection images in an image pattern on the one or more screens of the display device, the processor is configured to display, on the one or more screens of the display device, a pedestrian that is present on a right front side of the vehicle at the intersection and is crossing the second road in a direction approaching the vehicle in accordance with the image pattern, the processor is configured to display, on the one or more screens of the display device, a second vehicle that is present on a left front side of the vehicle at the intersection and is traveling in a direction away from the vehicle, and the processor is configured to receive vehicle information transmitted from the vehicle and present the vehicle information to a remote operator via the display device and a speaker, the vehicle information including surrounding situation information, sound detected by an in-vehicle microphone, object information related to a relative position and a relative speed of an object with respect to the vehicle, and vehicle state information including a speed, an acceleration, a yaw rate, and a steering angle of the vehicle.
 2. The display control device according to claim 1, wherein in the image process, the processor is configured to horizontally flip the one or more intersection images displayed on the one or more screens when the moving body enters the intersection from a second direction opposite to a first direction, with respect to the one or more intersection images displayed on the one or more screens when the moving body enters the intersection from the first direction.
 3. The display control device according to claim 1, wherein when the direction of the vehicle-traveling-direction component of the vector that indicates the imaging direction of each of the first and second cameras faces the traveling direction, the processor is configured, in the image process, to arrange, on the right side of the image of the second camera capturing the first area of the second road located on the left side of the intersection when viewed from the moving body, the image of the first camera capturing the second area of the second road located on the right side of the intersection, and to horizontally flip each of the images of the first and second cameras.
 4. A display control system, comprising: the display control device according to claim 1; and the one or more cameras, wherein the processor is configured to hide the one or more intersection images after the moving body has passed through the intersection.
-

